

MANOVA

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```
G1 = matrix(c(9,6,9,
              3,2,2), byrow=TRUE, nrow=2)
G2 = matrix(c(0,2,
              4,0), byrow=TRUE, nrow=2)
G3 = matrix(c(3,1,2,
              8,9,7),byrow = T, nrow=2)

p = 2
n1 = 3
n2 = 2
n3 = 3
n = n1 + n2 + n3
k = 3

X1 = G1
X2 = G2
X3 = G3
X = cbind(X1, X2, X3)

S1 = cov(t(X1));
S2 = cov(t(X2));
S3 = cov(t(X3));
S = cov(t(X));

A1 = (n1 - 1) * S1;
A2 = (n2 - 1) * S2;
A3 = (n3 - 1) * S3;
A = (n - 1) * S;

T_ = A
W = A1 + A2 + A3
B = T_ - W
cat("The within sum of square and cross product matrix is","\n")

## The within sum of square and cross product matrix is

print(W)

##      [,1]      [,2]
## [1,]    10 -4.000000
## [2,]    -4 10.666667

cat("The between matrix sum of square and cross product is","\n")

## The between matrix sum of square and cross product is
```

```

print(B)

##      [,1]      [,2]
## [1,]    78 -32.00000
## [2,]   -32  63.20833

L = det(W) / det(T_)
test_stat = -(n - 1 - ((p + k) / 2)) * log(L)
df = p * (k - 1)
p_value = pchisq(test_stat, df, lower.tail = FALSE)

cat("Wilks' Lambda:", L, "\n")
## Wilks' Lambda: 0.01741915

cat("Chi-square statistic:", test_stat, "\n")
## Chi-square statistic: 18.22583

cat("Degrees of freedom:", df, "\n")
## Degrees of freedom: 4

cat("P-value:", p_value, "\n")
## P-value: 0.001114774

```